

ONLINE APPENDIX: MONETARY POLICY AND SOVEREIGN DEBT SUSTAINABILITY

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Appendix A: An Economy with Costly Price Adjustment

In this appendix, we lay out a model economy with the following characteristics: (i) firms are explicitly modelled, (ii) a subset of them are price-setters but incur a convex cost for changing their nominal price, and (iii) the social welfare function and the equilibrium conditions are the same as in the model economy in the main text.

Final Good Producer

In the model laid out in the main text, we assumed that output of the single consumption good Y_t is exogenous. Consider now an alternative setup in which the single consumption good is produced by a representative, perfectly competitive final good producer with the following Dixit-Stiglitz technology,

$$Y_t = \left(\int_0^1 y_{it}^{(\varepsilon-1)/\varepsilon} di \right)^{\varepsilon/(\varepsilon-1)}, \quad (\text{A.1})$$

where $\{y_{it}\}$ is a continuum of intermediate goods and $\varepsilon > 1$. Let P_{it} denote the nominal price of intermediate good $i \in [0, 1]$. The firm chooses $\{y_{it}\}$ to maximize profits, $P_t Y_t - \int_0^1 P_{it} y_{it} di$, subject to (A.1). The first order conditions are

$$y_{it} = \left(\frac{P_{it}}{P_t} \right)^{-\varepsilon} Y_t, \quad (\text{A.2})$$

for each $i \in [0, 1]$. Assuming free entry, the zero profit condition and equations (A.2) imply $P_t = (\int_0^1 P_{it}^{1-\varepsilon} di)^{1/(1-\varepsilon)}$.

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Intermediate Goods Producers

Each intermediate good i is produced by a monopolistically competitive intermediate-good producer, which we will refer to as 'firm i ' henceforth for brevity. Firm i operates a linear production technology,

$$y_{it} = Z_t n_{it}, \quad (\text{A.3})$$

where n_{it} is labor input and $Z_t = e^{z_t}$ is productivity, where z_t follows equation (1) in the main text. At each point in time, firms can change the price of their product but face quadratic price adjustment cost as in Rotemberg (1982). Letting $\dot{P}_{it} \equiv dP_{it}/dt$ denote the change in the firm's price, price adjustment costs in units of the final good are given by

$$\Psi_t \left(\frac{\dot{P}_{it}}{P_{it}} \right) \equiv \frac{\psi(Y_t)}{2} \left(\frac{\dot{P}_{it}}{P_{it}} \right)^2 \tilde{C}_t, \quad (\text{A.4})$$

where $\tilde{C}_t$ is aggregate consumption. Let $\pi_{it} \equiv \dot{P}_{it}/P_{it}$ denote the rate of increase in the firm's price. The instantaneous profit function in units of the final good is given by

$$\begin{aligned} \Pi_{it} &= \frac{P_{it}}{P_t} y_{it} - w_t n_{it} - \Psi_t(\pi_{it}) \\ &= \left(\frac{P_{it}}{P_t} - \frac{w_t}{Z_t} \right) \left(\frac{P_{it}}{P_t} \right)^{-\varepsilon} Y_t - \Psi_t(\pi_{it}), \end{aligned} \quad (\text{A.5})$$

where w_t is the perfectly competitive real wage and in the second equality we have used (A.2) and (A.3). Without loss of generality, firms are assumed to be risk neutral and have the same discount factor as households, ρ . Then firm i 's objective function is

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t} \Pi_{it} dt,$$

with Π_{it} given by (A.5). Notice that the firm's optimization *problem* is not affected by sovereign defaults, although of course default does affect the aggregate variables that enter the firm's problem (Y_t , P_t , etc.). The state variable specific to firm i , P_{it} , evolves according to $dP_{it} = \pi_{it} P_{it} dt$. We conjecture that the aggregate state relevant to the firm's decisions can be summarized by $(b_t, z_t, P_t) \equiv S_t$.¹ Then firm i 's *value function* $J(P_{it}, S_t)$ must satisfy the following Hamilton-Jacobi-Bellman (HJB) equation,

$$\begin{aligned} \rho J(P_i, S) &= \max_{\pi_i} \left\{ \left(\frac{P_i}{P} - \frac{w}{e^z} \right) \left(\frac{P_i}{P} \right)^{-\varepsilon} Y - \Psi(\pi_i) + \pi_i P_i \frac{\partial J}{\partial P_i}(P_i, S) \right\} \\ &\quad + \mu'_S(S) D_S J(P_i, S) + \frac{1}{2} \sigma'_S(S) (D_{SS} J(P_i, S)) \sigma_S(S), \end{aligned}$$

1. In particular, we later show that in equilibrium $Y_t = Z_t$, whereas w_t and $\tilde{C}_t$ are also functions of (b_t, Z_t, P_t) . The states P_t and b_t follow the same laws of motion as in the main text, equations (2) and (4) respectively, whereas $z_t = \log Z_t$ follows equation (1).

where the vectors $(\mu_S(S), \sigma_S(S))$ collect the drift and diffusion terms, respectively, of the aggregate states S , and (D_S, D_{SS}) are the gradient and Hessian operators, respectively, with respect to S .² The first order and envelope conditions of this problem are (we omit the arguments of J to ease the notation),

$$\begin{aligned} \psi(Y) \pi_i \tilde{C} &= P_i \frac{\partial J}{\partial P_i}, \\ \rho \frac{\partial J}{\partial P_i} &= \left[\varepsilon \frac{w}{Z} - (\varepsilon - 1) \frac{P_i}{P} \right] \left(\frac{P_i}{P} \right)^{-\varepsilon} \frac{Y}{P_i} + \pi_i \left(\frac{\partial J}{\partial P_i} + P_i \frac{\partial^2 J}{\partial P_i^2} \right) \\ &\quad + \frac{\partial}{\partial P_i} \left[\mu'_S(S) D_S J + \frac{1}{2} \sigma'_S(S) (D_{SS} J) \sigma_S(S) \right]. \end{aligned}$$

In what follows, we will consider a symmetric equilibrium in which all firms choose the same price: $P_i = P$, $\pi_i = \pi$ for all i . After some algebra, it can be shown that the above conditions imply the following pricing Euler equation,³

$$\begin{aligned} \left(\rho - \frac{\tilde{C}_b(b, z) s(b, z) - \tilde{C}_z(b, z) \mu z}{\tilde{C}(b, z)} \right) \pi(b, z) &= \frac{\varepsilon - 1}{\psi} \left(\frac{\varepsilon}{\varepsilon - 1} \frac{w}{e^z} - 1 \right) \frac{e^z}{\tilde{C}(b, z)} \\ &\quad + s(b, z) \pi_b(b, z) - \mu z \pi_z(b, z) + \sigma^2 F(S). \end{aligned} \quad (\text{A.6})$$

where $\tilde{C}(b, z)$ and $\pi(b, z)$ denote the equilibrium policy functions for total spending and inflation, and $F(S)$ is a function of the aggregate state capturing the effect of aggregate uncertainty (σ) on firms' pricing decision. Equation (A.6) determines the market clearing wage w as a function of S .

Households and the Utility Costs of Inflation

The representative household's preferences are given by

$$\mathbb{E}_0 \int_0^\infty e^{-\rho t} \log(\tilde{C}_t) dt,$$

where $\tilde{C}_t$ is household consumption of the final good. Define *total real spending* as the sum of household consumption and price adjustment costs,

$$\begin{aligned} C_t &\equiv \tilde{C}_t + \int_0^1 \Psi_t(\pi_{it}) di \\ &= \tilde{C}_t + \frac{\psi(Y_t)}{2} \pi_t^2 \tilde{C}_t, \end{aligned} \quad (\text{A.7})$$

2. In particular, $\mu_S(S) = [s(b, z), -\mu z, \pi P]'$, where $s(b, z)$ is the drift of a as defined in section 2 of the main text; and $\sigma_S(S) = [0, \sigma, 0]'$.

3. The proof is available upon request.

where in the second equality we have used the definition of Ψ_t (equation A.4) and the symmetry across firms in equilibrium. Instantaneous utility can then be expressed as

$$\begin{aligned}\log(\tilde{C}_t) &= \log(C_t) - \log\left(1 + \frac{\psi(Y_t)}{2}\pi_t^2\right) \\ &= \log(C_t) - \frac{\psi(Y_t)}{2}\pi_t^2 + O\left(\left\|\frac{\psi(Y_t)}{2}\pi_t^2\right\|^2\right),\end{aligned}\quad (\text{A.8})$$

where $O(\|x\|^2)$ denotes terms of order second and higher in x . Expression (A.8) is the same as the utility function in the main text (equation 6), up to a first order approximation of $\log(1+x)$ around $x=0$, where $x \equiv \frac{\psi(Y_t)}{2}\pi_t^2$ represents the percentage of aggregate spending that is lost to price adjustment. For our baseline calibration, the latter object is relatively small even for relatively high inflation rates, and therefore so is the error in computing the utility losses from price adjustment. Therefore, the utility function in the main text (for $\gamma=1$) provides a fairly accurate approximation of the welfare losses caused by inflation in the economy with costly price adjustment described here. We have solved the equilibrium implied by the exact inflation disutility function in the first line of equation (A.8) and found that the results are virtually identical to those in the main text.

As in the model in the main text, the government rebates to the household all the net proceedings from its international credit operations, denoted by $\tilde{T}_t$ in nominal terms. We assume that the household supplies one unit of labor input inelastically: $n_t = 1$. It also receives firms' profits in a lump-sum manner. Thus the household's nominal budget constraint is

$$P_t \tilde{C}_t = P_t w_t + P_t \int_0^1 \Pi_{it} di + \tilde{T}_t.$$

In the symmetric equilibrium, each firm's labor demand is $n_{it} = y_{it}/Z_t = Y_t/Z_t$. Since labor supply equals one, labor market clearing requires

$$\int_0^1 n_{it} di = Y_t/Z_t = 1 \Leftrightarrow Y_t = Z_t.$$

Therefore, in equilibrium output is simply equal to exogenous productivity Z_t . Each firm's real profits equal $\Pi_{it} = Y_t - w_t - \frac{\psi(Y_t)}{2}\pi_t^2 \tilde{C}_t$. Using this in the household's budget constraint, we obtain

$$\tilde{T}_t = P_t \left(\tilde{C}_t + \frac{\psi(Y_t)}{2}\pi_t^2 \tilde{C}_t - Y_t \right) = P_t (C_t - Y_t),$$

where in the second equality we have used (A.7).

Fiscal and Monetary Policy

The government maximizes household welfare subject to the laws of motion of the aggregate state variables. The default scenario is the same as in the main text, with one

qualification: upon default and during the subsequent exclusion period, productivity equals $(Z_t - \varepsilon(Z_t))$. This, together with the fact that in equilibrium $Y_t = Z_t$, implies that the default scenario is exactly as in the main text. It is then trivial to show that the government's maximization problem is exactly the same as in the main text, once we take into account that (i) the welfare criterion is the same (equation 9), and (ii) the law of motion of the debt ratio is the same (equation 4). As a result, the policy functions for inflation and primary deficit ratio will also be the same: $\pi_t = \pi(b_t, z_t)$, $c_t = c(b_t, z_t)$.

Notice finally that, since $Y_t = Z_t$, in equilibrium we have $C_t = Z_t \equiv C(b_t, z_t)$, and therefore $\tilde{C}_t = C(b_t, z_t) / [1 + \frac{\psi(e^{z_t})}{2} \pi(b_t, z_t)^2] \equiv \tilde{C}(b_t, z_t)$. Likewise, the pricing Euler equation derived above (equation A.6) determines the market clearing wage given the aggregate state: $w_t = w(b_t, z_t, P_t)$. We thus verify our previous conjecture that (b_t, z_t, P_t) are the relevant aggregate states for firms.

Appendix B: Description of the Numerical Algorithm

The Augmented Model with an Option to Default

Here we present an algorithm to compute the equilibrium of the model with inflation. Our numerical algorithm is based on an augmented model which assumes that the government may only default when it receives an exogenous option to default. We assume that the option to default follows a Poisson process with parameter φ , that is, there is a number of random times $\{\tilde{t}_i\}_{i=1}^{\infty}$ at which the government decides whether to continue repaying its debt or to default. If the arrival rate tends to infinite $\varphi \rightarrow \infty$ the government can default at any point in time, whereas when $\varphi = 0$ default is not available. Therefore for a large enough value of φ this specification nests the case of continuous default discussed in the main body of the text. The advantage of this method is that it prevents some numerical problems associated with the numerical solution of default problems in continuous time discussed below. Instead of the level of debt, we employ the level of net assets $a = -b$.

The HJB in this case results in

$$\begin{aligned} \rho V(a, z) = & \max_{c, \pi, d \in \{0, 1\}} \frac{c^{1-\gamma} - 1}{1-\gamma} - \frac{\psi}{2} y^\xi \pi^2 + s(a, z, c, \pi) \frac{\partial V}{\partial a} - \mu z \frac{\partial V}{\partial z} \\ & + \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial z^2} + \varphi d(a, z) [V_{def}(a, z) - V(a, z)]. \end{aligned} \quad (B.1)$$

Notice that the variational inequality has been replaced by a HJB equation including the term

$$\varphi d(a, z) [V_{def}(a, z) - V(a, z)].$$

If the default policy d at the state (a, z) is one then the term indicates how in the case of the arrival of an option to default the value function jumps to the case with default, $V_{def}(a, z)$. The HJB for the default value (16) function remains the same.

Bond prices in this case are given by

$$(\bar{r} + \pi(a, z) + \lambda) Q(a, z) = (\lambda + \delta) + s(a, z) \frac{\partial Q}{\partial a} - \mu z \frac{\partial Q}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 Q}{\partial z^2} + \varphi d(a, z) [Q_{def}(a, z) - Q(a, z)], \quad (B.2)$$

where the term $\varphi d(a, z) [Q_{def}(a, z) - Q(a, z)]$ indicates that in the case of default the price of the bond is $Q_{def}(a, z)$. The advantage of this formulation, with a finite φ is that bond prices in the no-default region are never zero. There is a lower bound ($Q_{def}(a, z) = 0$)

$$Q^{\min}(a, z) = \frac{(\lambda + \delta)}{\bar{r} + \pi(a, z) + \lambda + \varphi} > 0,$$

which can be made arbitrarily small by increasing φ . Having a non-zero value of Q avoids the problem that the drift function $s(a, z)$ (defined in equation 11) becomes infinite at the default frontier as it is a function of the inverse of the bond price.

For values of φ larger than 4 – corresponding to an average default option per quarter – the equilibrium objects remain constant up to the considered precision. In any case, we calibrate $\varphi = 8$ to avoid any possible error associated with the numerical scheme.

Solution to the no Default Hamilton-Jacobi-Bellman Equation

The HJB equation is solved by a finite difference scheme following [Achdou et al. \(2020\)](#). It approximates the value function $V(a, z)$ on a finite grid with steps Δa and Δz : $a \in \{a_1, \dots, a_I\}$, $z \in \{z_1, \dots, z_J\}$. We use the notation $V_{i,j} := V(a_i, z_j)$, $i = 1, \dots, I$; $j = 1, \dots, J$. The derivative of V with respect to a can be approximated with either a forward or a backward approximation:

$$\frac{\partial V(a_i, z_j)}{\partial a} \approx \partial_{a,F} V_{i,j} := \frac{V_{i+1,j} - V_{i,j}}{\Delta a}, \quad (B.3)$$

$$\frac{\partial V(a_i, z_j)}{\partial a} \approx \partial_{a,B} V_{i,j} := \frac{V_{i,j} - V_{i-1,j}}{\Delta a}, \quad (B.4)$$

where the decision between one approximation or the other depends on the sign of the savings function $s_{i,j} = \left(\frac{\lambda + \delta}{Q(a_i, z_j)} - \lambda - \pi(a_i, z_j) \right) a_i + \frac{e^{z_j} - c_{i,j}}{Q(a_i, z_j)}$ through an “upwind scheme” described below. The derivative of V with respect to z is approximated using a forward approximation

$$\frac{\partial V(a_i, z_j)}{\partial z} \approx \partial_z V_{i,j} := \frac{V_{i,j+1} - V_{i,j}}{\Delta z}, \quad (B.5)$$

$$\frac{\partial^2 V(a_i, z_j)}{\partial z^2} \approx \partial_{zz} V_{i,j} := \frac{V_{i,j+1} + V_{i,j-1} - 2V_{i,j}}{(\Delta z)^2}. \quad (B.6)$$

The HJB equation (B.1) is

$$\begin{aligned} \rho V(a, z) = & u(c) - x(\pi, e^z) + \left[\left(\frac{\lambda + \delta}{Q(a, z)} - \lambda - \pi \right) a + \frac{e^z - c}{Q(a, z)} \right] \frac{\partial V}{\partial a} \\ & - \mu z \frac{\partial V}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial z^2} + \varphi d(a, z) [V_{def}(a, z) - V(a, z)], \end{aligned}$$

where

$$\begin{aligned} c &= (u')^{-1} \left[\frac{1}{Q(a, z)} \frac{\partial V}{\partial a} \right] = \left[\frac{1}{Q(a, z)} \frac{\partial V}{\partial a} \right]^{-1/\gamma} \\ \pi &= -\frac{a}{\psi e^{\xi z}} \frac{\partial V}{\partial a} = -\frac{a}{\psi e^{\xi z}} c^{-\gamma} Q(a, z). \end{aligned}$$

The HJB equation is approximated by an upwind scheme

$$\begin{aligned} \frac{V_{i,j}^{n+1} - V_{i,j}^n}{\Delta} + \rho V_{i,j}^{n+1} = & u(c_{i,j}^n) - x(\pi_{i,j}^n, e^{\xi z_j}) + \varphi d_{i,j}^n V_{def,i,j}^n \\ & + \partial_{a,F} V_{i,j}^{n+1} s_{i,j,F}^n \mathbf{1}_{s_{i,j,F}^n > 0} \\ & + \partial_{a,B} V_{i,j}^{n+1} s_{i,j,B}^n \mathbf{1}_{s_{i,j,B}^n < 0} - \mu z_j \partial_z V_{i,j}^{n+1} + \frac{\sigma_z^2}{2} \partial_{zz} V_{i,j}^{n+1} - \varphi d_{i,j}^n V_{i,j}^{n+1}, \end{aligned}$$

where

$$\begin{aligned} s_{i,j,F}^n &= \left[\left(\frac{\lambda + \delta}{Q_{i,j}^n} - \lambda - \frac{-a_i}{\psi} \partial_{a,F} V_{i,j}^n \right) a_i + \frac{e^{z_j} - (u')^{-1} \left[\frac{\partial_{a,F} V_{i,j}^n}{Q_{i,j}^n} \right]}{Q_{i,j}^n} \right], \\ s_{i,j,B}^n &= \left[\left(\frac{\lambda + \delta}{Q_{i,j}^n} - \lambda - \frac{-a_i}{\psi} \partial_{a,B} V_{i,j}^n \right) a_i + \frac{e^{z_j} - (u')^{-1} \left[\frac{\partial_{a,B} V_{i,j}^n}{Q_{i,j}^n} \right]}{Q_{i,j}^n} \right]. \end{aligned}$$

This can be expressed as:

$$\begin{aligned} \frac{V_{i,j}^{n+1} - V_{i,j}^n}{\Delta} + \rho V_{i,j}^{n+1} = & u(c_{i,j}^n) - x(\pi_{i,j}^n, e^{\xi z_j}) + \varphi d_{i,j}^n V_{def,i,j}^n + V_{i-1,j}^{n+1} q_{i,j} \\ & + V_{i,j}^{n+1} \beta_{i,j} + V_{i+1,j}^{n+1} \eta_{i,j} + V_{i,j-1}^{n+1} \xi + V_{i,j+1}^{n+1} \varsigma_j, \quad (\text{B.7}) \end{aligned}$$

where

$$c_{i,j}^n = (u')^{-1} \left[\frac{1}{Q_{i,j}^n} \left(\partial_{a,F} V_{i,j}^n \mathbf{1}_{s_{i,j,F}^n > 0} + \partial_{a,B} V_{i,j}^n \mathbf{1}_{s_{i,j,B}^n < 0} \right. \right. \\ \left. \left. + Q_{i,j}^n u' \left(c_{i,j}^{0,n} \right) \mathbf{1}_{s_{i,j,F}^n < 0, s_{i,j,B}^n > 0} \right) \right], \quad (\text{B.8})$$

$$\pi_{i,j}^n = -\frac{a_i}{\psi e^{\xi z_j}} (c_{i,j}^n)^{-\gamma} Q_{i,j}^n, \quad (\text{B.9})$$

$$Q_{i,j} = -\frac{s_{i,j,B}^n \mathbf{1}_{s_{i,j,B}^n < 0}}{\Delta a}, \\ \beta_{i,j} = -\frac{s_{i,j,F}^n \mathbf{1}_{s_{i,j,F}^n > 0}}{\Delta a} + \frac{s_{i,j,B}^n \mathbf{1}_{s_{i,j,B}^n < 0}}{\Delta a} + \frac{\mu z_j}{\Delta z} - \frac{\sigma^2}{(\Delta z)^2} - d_{i,j}^n \varphi, \\ \eta_{i,j} = \frac{s_{i,j,F}^n \mathbf{1}_{s_{i,j,F}^n > 0}}{\Delta a}, \\ \xi = \frac{\sigma^2}{2(\Delta z)^2}, \\ \varsigma_j = \frac{\sigma^2}{2(\Delta z)^2} - \frac{\mu z_j}{\Delta z},$$

where

$$c_{i,j}^{0,n} = Q_{i,j}^n \left(\frac{\lambda + \delta}{Q(a_i, z_j)} - \lambda - \pi_{i,j}^{0,n} \right) a_i + e^{z_j}, \\ \pi_{i,j}^{0,n} = \left(\frac{-a}{\psi, e^{\xi z_j}} \partial_{a,F} V_{i,j}^n + \frac{-a}{\psi, e^{\xi z_j}} \partial_{a,B} V_{i,j}^n \right) / 2.$$

The state constraint $a \leq 0$ is enforced by setting $s_{i,j,F}^n = 0$. Similarly, we impose $a \geq a^{\min}$ for a value of $a^{\min}$ large enough, which requires $s_{1,j,B}^n = 0$. Therefore, the values $V_{0,j}^{n+1}$ and $V_{I+1,j}^{n+1}$ are never used. The boundary conditions with respect to z are

$$\frac{\partial V(a, \underline{z})}{\partial z} = \frac{\partial V(a, \bar{z})}{\partial z} = 0,$$

as the process is reflected. At the boundaries in the j dimension, equation (B.7) becomes

$$\frac{V_{i,1}^{n+1} - V_{i,1}^n}{\Delta} + \rho V_{i,j}^{n+1} = u(c_{i,1}^n) - x(\pi_{i,j}^n) + \varphi d_{i,1}^n V_{def,i,1}^n \\ + V_{i-1,1}^{n+1} Q_{i,1} + V_{i,1}^{n+1} (\beta_{i,1} + \xi) + V_{i+1,1}^{n+1} \eta_{i,1} + V_{i,2}^{n+1} \varsigma_1, \\ \frac{V_{i,J}^{n+1} - V_{i,J}^n}{\Delta} + \rho V_{i,j}^{n+1} = u(c_{i,J}^n) - x(\pi_{i,j}^n) + \varphi d_{i,J}^n V_{def,i,J}^n \\ + V_{i-1,J}^{n+1} Q_{i,J} + V_{i,J}^{n+1} (\beta_{i,J} + \varsigma_J) + V_{i+1,J}^{n+1} \eta_{i,J} + V_{i,J-1}^{n+1} \xi_J.$$

Equation (B.7) is a system of $I \times J$ linear equations which can be written in matrix notation as:

$$\frac{\mathbf{V}^{n+1} - \mathbf{V}^n}{\Delta} + \rho \mathbf{V}^{n+1} = \mathbf{u}^n + \mathbf{A}^n \mathbf{V}^{n+1}, \quad (\text{B.10})$$

where the matrix $\mathbf{A}^n$ and the vectors $\mathbf{V}^{n+1}$ and $\mathbf{u}^n$ are defined by:

$$\mathbf{A}^n = \begin{bmatrix} \beta_{1,1} + \xi & \eta_{1,1} & 0 & \cdots & 0 & \varsigma_1 & 0 & 0 & \cdots & 0 \\ \varrho_{2,1} & \beta_{2,1} + \xi & \eta_{2,1} & \cdots & 0 & 0 & \varsigma_1 & 0 & \cdots & 0 \\ 0 & \varrho_{3,1} & \beta_{3,1} + \xi & \eta_{3,1} & 0 & \cdots & 0 & \varsigma_1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \varrho_{I,1} & \beta_{I,1} + \xi & \eta_{I,1} & 0 & 0 & \cdots & 0 \\ \xi & 0 & \cdots & 0 & \varrho_{1,2} & \beta_{1,2} & \eta_{1,2} & 0 & \cdots & 0 \\ 0 & \xi & \cdots & 0 & 0 & \varrho_{2,2} & \beta_{2,2} & \eta_{2,2} & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & \varrho_{I-1,J} & \beta_{I-1,J} + \varsigma_J & \eta_{I-1,J} \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \varrho_{I,J} & \beta_{I,I} + \varsigma_J \end{bmatrix}, \quad (\text{B.11})$$

$$\mathbf{V}^{n+1} = \begin{bmatrix} V_{1,1}^{n+1} \\ V_{2,1}^{n+1} \\ \vdots \\ V_{1,2}^{n+1} \\ V_{2,2}^{n+1} \\ \vdots \\ V_{I-1,J}^{n+1} \\ V_{I,J}^{n+1} \end{bmatrix}, \quad \mathbf{u}^n = \begin{bmatrix} u(c_{1,1}^n) - x(\pi_{1,1}^n, e^{z_1}) + \varphi d_{1,1}^n V_{def,1,1}^n \\ u(c_{2,1}^n) - x(\pi_{2,1}^n, e^{z_1}) + \varphi d_{2,1}^n V_{def,2,1}^n \\ \vdots \\ u(c_{1,2}^n) - x(\pi_{1,2}^n, e^{z_2}) + \varphi d_{1,2}^n V_{def,1,2}^n \\ u(c_{2,2}^n) - x(\pi_{2,2}^n, e^{z_2}) + \varphi d_{2,2}^n V_{def,2,2}^n \\ \vdots \\ u(c_{I-1,J}^n) - x(\pi_{I-1,J}^n, e^{z_J}) + \varphi d_{I-1,J}^n V_{def,I-1,J}^n \\ u(c_{I,J}^n) - x(\pi_{I,J}^n, e^{z_J}) + \varphi d_{I,J}^n V_{def,I,J}^n \end{bmatrix}.$$

The system can in turn be written as

$$\mathbf{B}^n \mathbf{V}^{n+1} = \mathbf{d}^n,$$

where $\mathbf{B}^n = (\frac{1}{\Delta} + \rho) \mathbf{I} - \mathbf{A}^n$ and $\mathbf{d}^n = \mathbf{u}^n + \frac{\mathbf{V}^n}{\Delta}$. $\mathbf{I}$ is the identity matrix. Matrix $\mathbf{B}^n$ is a sparse matrix, and the system can be efficiently solved in Matlab.

In order to avoid abrupt jumps in bond prices we combine it with a relaxation scheme such that, given a constant $\kappa \in (0, 1)$, if we denote the result of the linear system of equations (B.10) as $\hat{V}_{i,j}^{n+1}$, then

$$V_{i,j}^{n+1} = \kappa \hat{V}_{i,j}^{n+1} + (1 - \kappa) V_{i,j}^n. \quad (\text{B.12})$$

Solution to the Default Hamilton-Jacobi-Bellman Equation (no Inflation)

The HJB equation in this case (16) with $\pi = 0$ is

$$\rho V_{def}(a, z) = u_{def}(z) - \mu z \frac{\partial V_{def}}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V_{def}}{\partial z^2} + \chi [V(\theta a, z) - V_{def}(a, z)],$$

which, using a finite difference method identical to the one described above (this time no upwind is necessary as there is no control), yields

$$\frac{V_{def,i,j}^{n+1} - V_{def,i,j}^n}{\Delta} + \rho V_{i,j}^{n+1} = u_{def}(z_j) + \chi V_{\theta i,j}^n + V_{def,i,j}^{n+1} \hat{\beta}_j + V_{def,i,j-1}^{n+1} \xi + V_{def,i,j+1}^{n+1} \varsigma_j,$$

where $\hat{\beta}_j = \frac{\mu z_j}{\Delta z} - \frac{\sigma^2}{(\Delta z)^2} - \chi$. $V_{\theta i,j}^n$ is the interpolated value corresponding to the notional grid point θi . This equation can be written in matrix notation as:

$$\frac{\mathbf{V}_{def}^{n+1} - \mathbf{V}_{def}^n}{\Delta} + \rho \mathbf{V}_{def}^{n+1} = \mathbf{u}_{def}^n + \mathbf{A}_{def}^n \mathbf{V}_{def}^{n+1}, \quad (\text{B.13})$$

where

$$\mathbf{A}_{def}^n = \begin{bmatrix} \beta_1 + \xi & 0 & 0 & \cdots & 0 & \varsigma_1 & 0 & 0 & \cdots & 0 \\ 0 & \beta_1 + \xi & 0 & 0 & \cdots & 0 & \varsigma_1 & 0 & \cdots & 0 \\ 0 & 0 & \beta_1 + \xi & 0 & 0 & \cdots & 0 & \varsigma_1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & \beta_1 + \xi & 0 & 0 & 0 & \cdots & 0 \\ \xi & 0 & \cdots & 0 & 0 & \beta_2 & 0 & 0 & \cdots & 0 \\ 0 & \xi & \cdots & 0 & 0 & 0 & \beta_2 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \beta_J + \varsigma_J & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \beta_J + \varsigma_J \end{bmatrix},$$

$$\mathbf{V}_{def}^{n+1} = \begin{bmatrix} V_{def,1,1}^{n+1} \\ V_{def,2,1}^{n+1} \\ \vdots \\ V_{def,1,2}^{n+1} \\ V_{def,2,2}^{n+1} \\ \vdots \\ V_{def,I-1,J}^{n+1} \\ V_{def,I,J}^{n+1} \end{bmatrix}, \quad \mathbf{u}_{def}^n = \begin{bmatrix} u_{def}(z_1) + \chi V_{\theta 1,1}^n \\ u_{def}(z_1) + \chi V_{\theta 2,1}^n \\ \vdots \\ u_{def}(z_2) + \chi V_{\theta 1,2}^n \\ u_{def}(z_2) + \chi V_{\theta 2,2}^n \\ \vdots \\ u_{def}(z_J) + \chi V_{\theta(I-1),J}^n \\ u_{def}(z_J) + \chi V_{\theta I,J}^n \end{bmatrix}. \quad (\text{B.14})$$

Solution to the Default Hamilton-Jacobi-Bellman Equation (Optimal Inflation)

The HJB equation in this case (16) with optimal inflation is

$$\begin{aligned} \rho V_{def}(a, z) = & u_{def}(z) - x(\pi, e^z - \varepsilon(e^z)) - \pi a \frac{\partial V_{def}}{\partial a} \\ & - \mu z \frac{\partial V_{def}}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V_{def}}{\partial z^2} + \chi [V(\theta a, z) - V_{def}(a, z)], \end{aligned}$$

with a first order condition $\pi = -\frac{a}{\psi(e^z - \varepsilon(e^z))} \frac{\partial V_{def}}{\partial a}$. Proceeding as in the case without default, we construct a matrix $\mathbf{A}_{def}^n$ equal to matrix $\mathbf{A}^n$ in (B.11) except for the fact that

the drift now is

$$s_{def,i,j,F}^n = \frac{a_i^2}{\psi(e^{\xi z_j} - \varepsilon(e^{\xi z_j}))} \partial_{a,F} V_{def,i,j}^n,$$

$$s_{def,i,j,B}^n = \frac{a_i^2}{\psi(e^{\xi z_j} - \varepsilon(e^{\xi z_j}))} \partial_{a,B} V_{def,i,j}^n,$$

and that element $\beta_{i,j} = -\frac{s_{i,j,F}^n \mathbf{1}_{s_{i,j,F}^n > 0}}{\Delta a} + \frac{s_{i,j,B}^n \mathbf{1}_{s_{i,j,B}^n < 0}}{\Delta a} + \frac{\mu z_j}{\Delta z} - \frac{\sigma^2}{(\Delta z)^2} - \chi$.

Vectors $\mathbf{V}_{def}^{n+1}$ and $\mathbf{u}_{def}^n$ are defined as in (15) except for the fact that utility is now $u_{def}(z_j) - x \left(\pi_{def,i,j}^n, e^{\xi z_j} \right) + \chi V_{\theta i,j}^n$ with

$$\pi_{def,i,j}^n = -\frac{a_i}{\psi(e^{\xi z_j} - \varepsilon(e^{\xi z_j}))} \left(\partial_{a,F} V_{def,i,j}^n \mathbf{1}_{s_{i,j,F}^n > 0} + \partial_{a,B} V_{i,j}^n \mathbf{1}_{s_{i,j,B}^n < 0} \right).$$

Solution to the Bond Price Equation

The bond price equation (18) is

$$Q(a, z) (\bar{r} + \lambda + \pi(a, z)) = (\lambda + \delta) + s(a, z) \frac{\partial Q}{\partial a} \\ - \mu z \frac{\partial Q}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 Q}{\partial z^2} + \varphi \mathbf{1}_{d(a,z) > 0} [Q_{def}(a, z) - Q(a, z)]$$

which, using again a finite difference method identical to the ones already described above (this time no upwind is necessary either), yields

$$\frac{Q_{i,j}^{n+1} - Q_{i,j}^n}{\Delta} + (\bar{r} + \lambda + \pi_{i,j}^n) Q_{i,j}^{n+1} = (\lambda + \delta) + \varphi \mathbf{1}_{d_{i,j} > 0} Q_{def,i,j}^n + Q_{i-1,j}^{n+1} \varrho_{i,j} \\ + Q_{i,j}^{n+1} \beta_{i,j} + Q_{i+1,j}^{n+1} \chi_{i,j} + Q_{i,j-1}^{n+1} \xi_j + Q_{i,j+1}^{n+1} \varsigma_j,$$

where

$$\frac{\mathbf{Q}^{n+1} - \mathbf{Q}^n}{\Delta} + (\bar{r} + \lambda + \dots^n) \mathbf{Q}^{n+1} = \tilde{\mathbf{Q}}_{def}^{n+1} + \mathbf{A}^n \mathbf{Q}^{n+1}, \quad (\text{B.15})$$

where $\dots^n$ is a diagonal matrix with elements $\pi_{i,j}^n$ and

$$\mathbf{Q}^{n+1} = \begin{bmatrix} Q_{1,1}^{n+1} \\ Q_{2,1}^{n+1} \\ \vdots \\ Q_{1,2}^{n+1} \\ Q_{2,2}^{n+1} \\ \vdots \\ Q_{I-1,J}^{n+1} \\ Q_{I,J}^{n+1} \end{bmatrix}, \quad \tilde{\mathbf{Q}}_{def}^{n+1} = \begin{bmatrix} (\lambda + \delta) + \varphi \mathbf{1}_{d_{1,1} > 0} Q_{def,1,1}^{n+1} \\ (\lambda + \delta) + \varphi \mathbf{1}_{d_{2,1} > 0} Q_{def,2,1}^{n+1} \\ \vdots \\ (\lambda + \delta) + \varphi \mathbf{1}_{d_{1,2} > 0} Q_{def,1,2}^{n+1} \\ (\lambda + \delta) + \varphi \mathbf{1}_{d_{2,2} > 0} Q_{def,2,2}^{n+1} \\ \vdots \\ (\lambda + \delta) + \varphi \mathbf{1}_{d_{I-1,J} > 0} Q_{def,I-1,J}^{n+1} \\ (\lambda + \delta) + \varphi \mathbf{1}_{d_{I,J} > 0} Q_{def,I,J}^{n+1} \end{bmatrix}.$$

We also combine it with a relaxation scheme such that, if we denote the result of the linear system of equations (B.15) as $\hat{Q}_{i,j}^{n+1}$, then

$$Q_{i,j}^{n+1} = \kappa \hat{Q}_{i,j}^{n+1} + (1 - \kappa) Q_{i,j}^n. \quad (\text{B.16})$$

Solution to the Default Bond Price Equation (no Inflation)

The bond pricing equation in this case is

$$\bar{r} Q_{def}(a, z) = -\mu z \frac{\partial Q_{def}}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 Q_{def}}{\partial z^2} + \chi [\theta Q(\theta a, z) - Q_{def}(a, z)],$$

which, using a finite difference method identical to the one described above (this time no upwind is necessary as there is no control), yields

$$\frac{Q_{def,i,j}^{n+1} - Q_{def,i,j}^n}{\Delta} + \bar{r} Q_{i,j}^{n+1} = \chi \theta Q_{\theta i,j}^n + Q_{def,i,j}^{n+1} \hat{\beta}_j + Q_{def,i,j-1}^{n+1} \xi + Q_{def,i,j+1}^{n+1} \varsigma_j.$$

$Q_{\theta i,j}^n$ is the interpolated value corresponding to the notional grid point θi . This equation can be written in matrix notation as:

$$\frac{\mathbf{Q}_{def}^{n+1} - \mathbf{Q}_{def}^n}{\Delta} + \bar{r} \mathbf{Q}_{def}^{n+1} = \mathbf{q}_{def}^n + \mathbf{A}_{def}^n \mathbf{Q}_{def}^{n+1}, \quad (\text{B.17})$$

where

$$\mathbf{Q}_{def}^{n+1} = \begin{bmatrix} Q_{def,1,1}^{n+1} \\ Q_{def,2,1}^{n+1} \\ \vdots \\ Q_{def,1,2}^{n+1} \\ Q_{def,2,2}^{n+1} \\ \vdots \\ Q_{def,I-1,J}^{n+1} \\ Q_{def,I,J}^{n+1} \end{bmatrix}, \quad \mathbf{q}_{def}^n = \begin{bmatrix} \chi \theta Q_{\theta 1,1}^n \\ \chi \theta Q_{\theta 2,1}^n \\ \vdots \\ \chi \theta Q_{\theta 1,2}^n \\ \chi \theta Q_{\theta 2,2}^n \\ \vdots \\ \chi \theta Q_{\theta(I-1),J}^n \\ \chi \theta Q_{\theta I,J}^n \end{bmatrix}.$$

Solution to the Default Bond Price Equation (Optimal Inflation)

The bond pricing equation in this case is

$$(\bar{r} + \pi(a, z)) Q_{def}(b, z) = -\pi a \frac{\partial Q_{def}}{\partial a} - \mu z \frac{\partial Q_{def}}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 Q_{def}}{\partial z^2} + \chi [\theta Q(\theta a, z) - Q_{def}(a, z)].$$

This equation can be written in matrix notation as:

$$\frac{\mathbf{Q}_{def}^{n+1} - \mathbf{Q}_{def}^n}{\Delta} + (\bar{r} + \dots^n) \mathbf{Q}_{def}^{n+1} = \mathbf{q}_{def}^n + \mathbf{A}_{def}^n \mathbf{Q}_{def}^{n+1}, \quad (\text{B.18})$$

where matrix $\mathbf{A}_{def}^n$ is the one constructed in the computation of the value function with default above.

Complete Algorithm

The algorithm to solve the model is based on two loops, an inner loop that finds the value functions and bond prices given the optimal default, and an outer one which computes the optimal default.

Outer Loop. Begin with an initial guess $V_{i,j}^0 = u(rai + z_j)/\rho$, $V_{def,i,j}^0 = [u_{def}(1) + \chi V_{i,j}^0]/(\rho + \chi)$, $Q_{i,j}^0 = 1$, $d_{i,j} = 0$ and set $m = 0$. Then:

1. Given $d_{i,j}^m$, run the inner loop to find $V_{i,j}^{m+1}$, $V_{def,i,j}^{m+1}$, $Q_{i,j}^{m+1}$ using as an initial guess $V_{i,j}^m$, $V_{def,i,j}^m$, $Q_{i,j}^m$, $Q_{def,i,j}^m$.
2. Compute $d_{i,j}^{m+1} = 1_{V_{def,i,j}^{m+1} > V_{i,j}^{m+1}}$ according to (14).
3. If $d_{i,j}^{m+1}$ is close enough to $d_{i,j}^m$ stop. If not set $m := m + 1$ and go to step 1.

Inner Loop. Given a default policy $d_{i,j}^m = 0$, begin with an initial guess $V_{i,j}^0 = V_{i,j}^m$, $V_{def,i,j}^0 = V_{def,i,j}^m$, $Q_{i,j}^0 = Q_{i,j}^m$ and set $n = 0$. Then:

1. Compute $\partial_{a,F} V_{i,j}^n$, $\partial_{a,B} V_{i,j}^n$, $\partial_z V_{i,j}^n$ and $\partial_{zz} V_{i,j}^n$ using (B.3)-(B.6).
2. Compute $c_{i,j}^n$ and $\pi_{i,j}^n$ using (B.8) and (B.9), respectively.
3. Find $V_{i,j}^{n+1}$ solving the linear system of equations (B.10) plus the relaxation scheme (B.12).
4. Find $V_{def,i,j}^{n+1}$ solving the linear system of equations (B.13).
5. Find $Q_{i,j}^{n+1}$ solving the linear system of equations (B.15) plus the relaxation scheme (B.16).
6. If $V_{i,j}^{n+1}$ is close enough to $V_{i,j}^n$ and $Q_{i,j}^{n+1}$ is close enough to $Q_{i,j}^n$, stop. If not set $n := n + 1$ and go to step 1.

Solution to the Kolmogorov Forward Equation

Finally, we describe here the algorithm to solve the Kolmogorov Forward equation

$$0 = -\frac{\partial}{\partial a} [s(a, z) g(a, z)] + \frac{\partial}{\partial z} [\mu z g(a, z)] + \frac{\sigma^2}{2} \frac{\partial^2 g}{\partial z^2} - \varphi d(a, z) g(a, z) + \frac{\chi}{\theta} g^{def}(a/\theta, z), \quad (\text{B.19})$$

$$0 = -\frac{\partial}{\partial a} [s^{def}(a, z) g^{def}(a, z)] + \frac{\partial}{\partial z} [\mu z g^{def}(a, z)] + \frac{\sigma^2}{2} \frac{\partial^2 g^{def}}{\partial z^2} + \varphi d(a, z) g(a, z) - \chi g^{def}(a, z), \quad (\text{B.20})$$

where $s^{def}(a, z) \equiv -\pi a$. We solve the equation using an upwind finite difference scheme. We use the notation $g_{i,j} \equiv g(a_i, z_j)$. The solution in matrix form is

$$\begin{aligned} \mathbf{A}^T \mathbf{g} + \frac{\chi}{\theta} \mathbf{g}_\theta^{def} &= 0, \\ \mathbf{A}_{def}^T \mathbf{g}^{def} + \varphi \mathbf{D} \mathbf{g} &= 0, \end{aligned}$$

where $\mathbf{g}_{\theta}^{def}$ is the matrix corresponding to the interpolated versions of $g_{\theta,i,j}^{def} = g^{def}(a_i/\theta, z_j)$ and

$$\mathbf{D}^n = \begin{bmatrix} d_{1,1} & 0 & 0 & \cdots & 0 & 0 \\ 0 & d_{2,1} & 0 & 0 & \cdots & 0 \\ 0 & 0 & d_{3,1} & 0 & 0 & \cdots \\ \vdots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & 0 & \cdots & 0 & d_{I-1,J} & 0 \\ 0 & 0 & \cdots & 0 & 0 & d_{I,J} \end{bmatrix}.$$

We solve the system iteratively using a standard linear equation solver. Finally, we normalize the joint distribution $g + g^{def}$ to one.

Appendix C: Welfare Decomposition

We first show how one can express the two components of the welfare decomposition (equations 22 and 23) recursively. We start by expressing the HJB equation of the repayment value function (equation B.1) as

$$[\rho + \varphi d(b, z)] V(b, z) = u(c(b, z)) - \frac{\psi}{2} \pi(b, z)^2 + s(b, z) \frac{\partial V}{\partial b} - \mu z \frac{\partial V}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V}{\partial z^2} + \varphi d(b, z) V_{def}(z).$$

Similarly, we express the HJB equation of the default value function (equation 16) as

$$(\rho + \chi) V_{def}(z) = u_{def}(z) - \mu z \frac{\partial V_{def}}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V_{def}}{\partial z^2} + \chi V(\theta a, z).$$

Using our postulated decomposition, $V(b, z) = V_c(b, z) + V_{\pi}(b, z)$ and $V_{def}(b, z) = V_{def,c}(b, z) + V_{def,\pi}(b, z)$, in the above two expressions, and ignoring function arguments except where needed, we obtain

$$\begin{aligned} [\rho + \varphi d](V_c + V_{\pi}) &= u(c) - \frac{\psi}{2} \pi^2 + s \left(\frac{\partial V_c}{\partial b} + \frac{\partial V_{\pi}}{\partial b} \right) - \mu z \left(\frac{\partial V_c}{\partial z} + \frac{\partial V_{\pi}}{\partial z} \right) \\ &\quad + \frac{\sigma^2}{2} \left(\frac{\partial^2 V_c}{\partial z^2} + \frac{\partial^2 V_{\pi}}{\partial z^2} \right) + \varphi d (V_{def,c} + V_{def,\pi}), \end{aligned}$$

$$\begin{aligned} (\rho + \chi) (V_{def,c} + V_{def,\pi}) &= u_{def} - \mu z \left(\frac{\partial V_{def,c}}{\partial z} + \frac{\partial V_{def,\pi}}{\partial z} \right) + \frac{\sigma^2}{2} \left(\frac{\partial^2 V_{def,c}}{\partial z^2} + \frac{\partial^2 V_{def,\pi}}{\partial z^2} \right) \\ &\quad + \chi [V_c(\theta a, z) + V_{\pi}(\theta a, z)]. \end{aligned}$$

We can then write

$$[\rho + \varphi d] V_c = u(c) + s \frac{\partial V_c}{\partial b} - \mu z \frac{\partial V_c}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V_c}{\partial z^2} + \varphi d V_{def,c},$$

$$\begin{aligned}
[\rho + \varphi d] V_\pi &= -\frac{\psi}{2} \pi^2 + s \frac{\partial V_\pi}{\partial b} - \mu z \frac{\partial V_\pi}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V_\pi}{\partial z^2} + \varphi d V_{def,\pi}, \\
(\rho + \chi) V_{def,c} &= u_{def} - \mu z \frac{\partial V_{def,c}}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V_{def,c}}{\partial z^2} + \chi V_c(\theta a, z), \\
(\rho + \chi) V_{def,\pi} &= 0 - \mu z \frac{\partial V_{def,\pi}}{\partial z} + \frac{\sigma^2}{2} \frac{\partial^2 V_{def,\pi}}{\partial z^2} + \chi V_\pi(\theta a, z).
\end{aligned}$$

These four value functions can be then be solved using finite-difference methods similar to those described in Appendix B.

Appendix D: Additional Figures

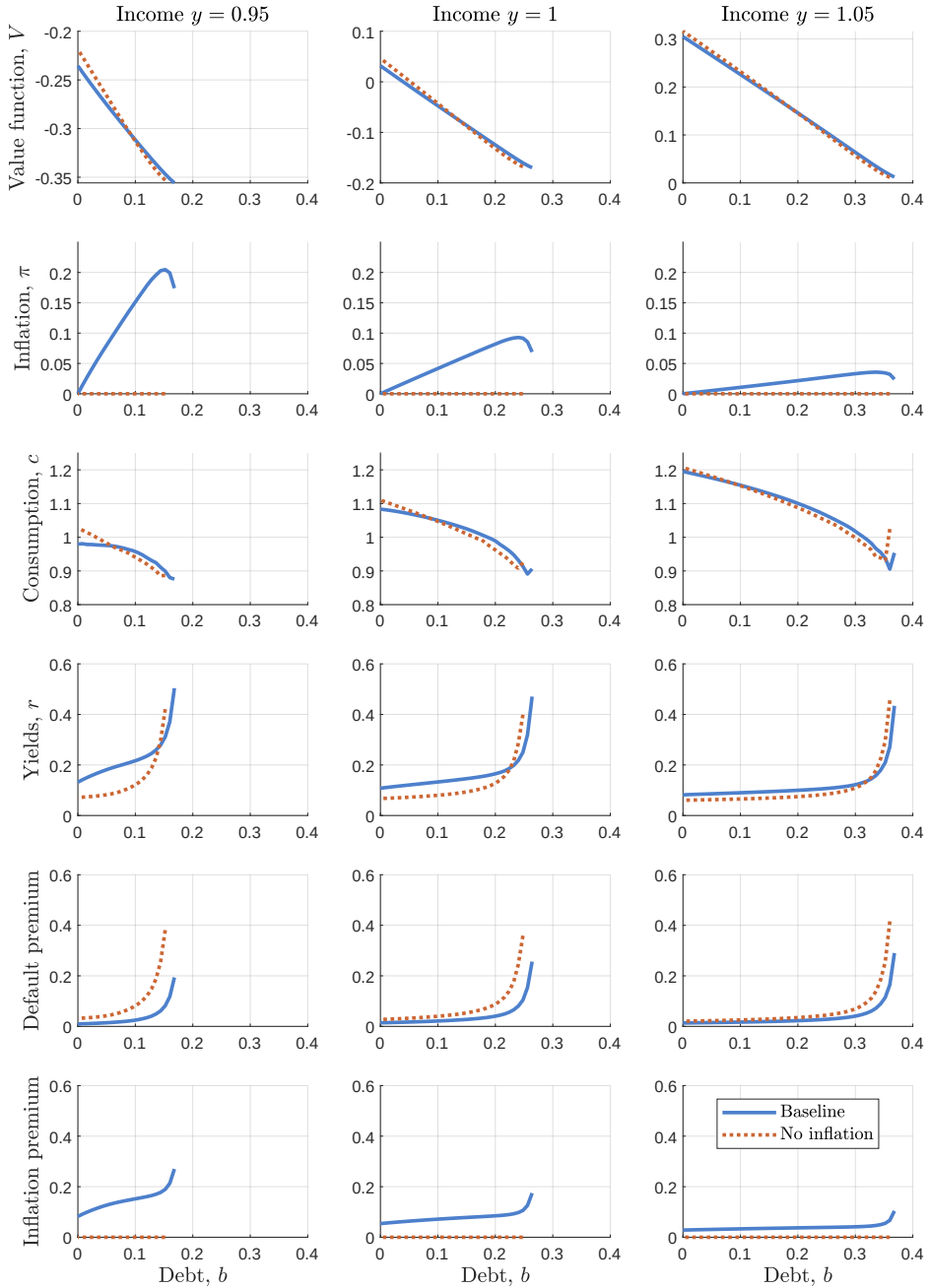


FIGURE D.1. Equilibrium objects. The figure shows the equilibrium objects in the repayment (thick line) and default (thin line) segments of debt for different values y .

Convergence of the policy function for Q , with different options for the initial guess

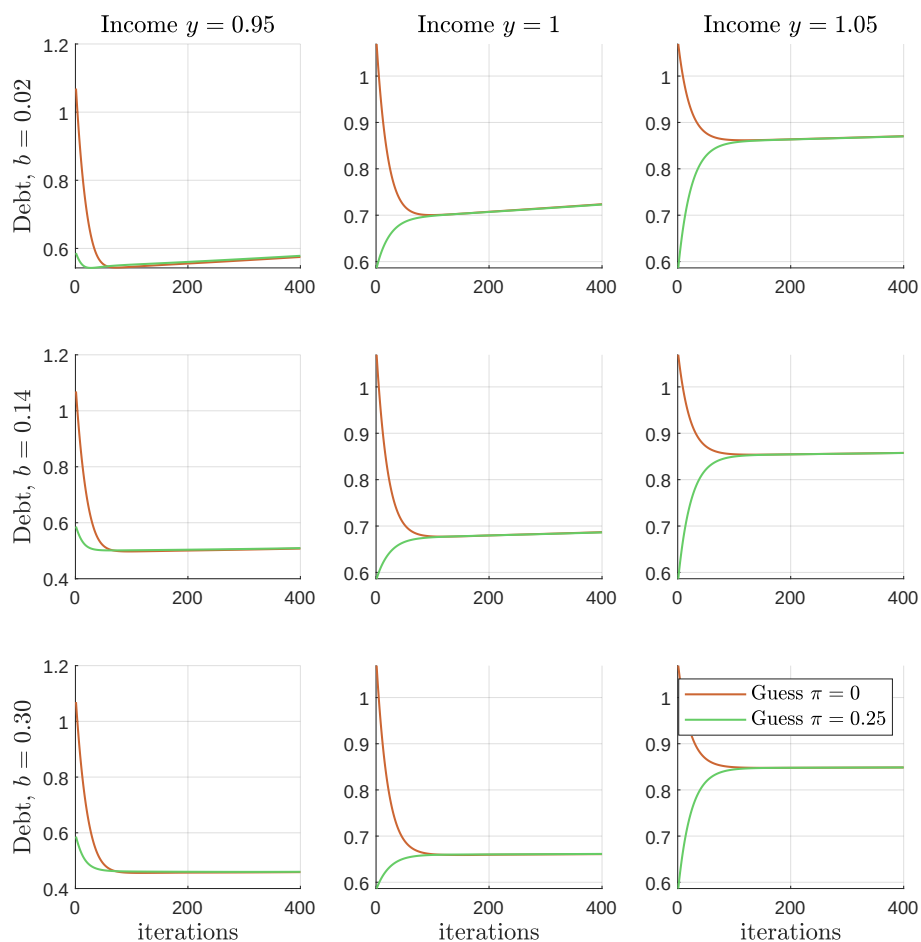


FIGURE D.2. Convergence across different initial guesses. The figure shows the convergence of two initial guesses of the bond price schedule, consistent with inflation levels of $\pi = 0$ and $\pi = 0.25$.

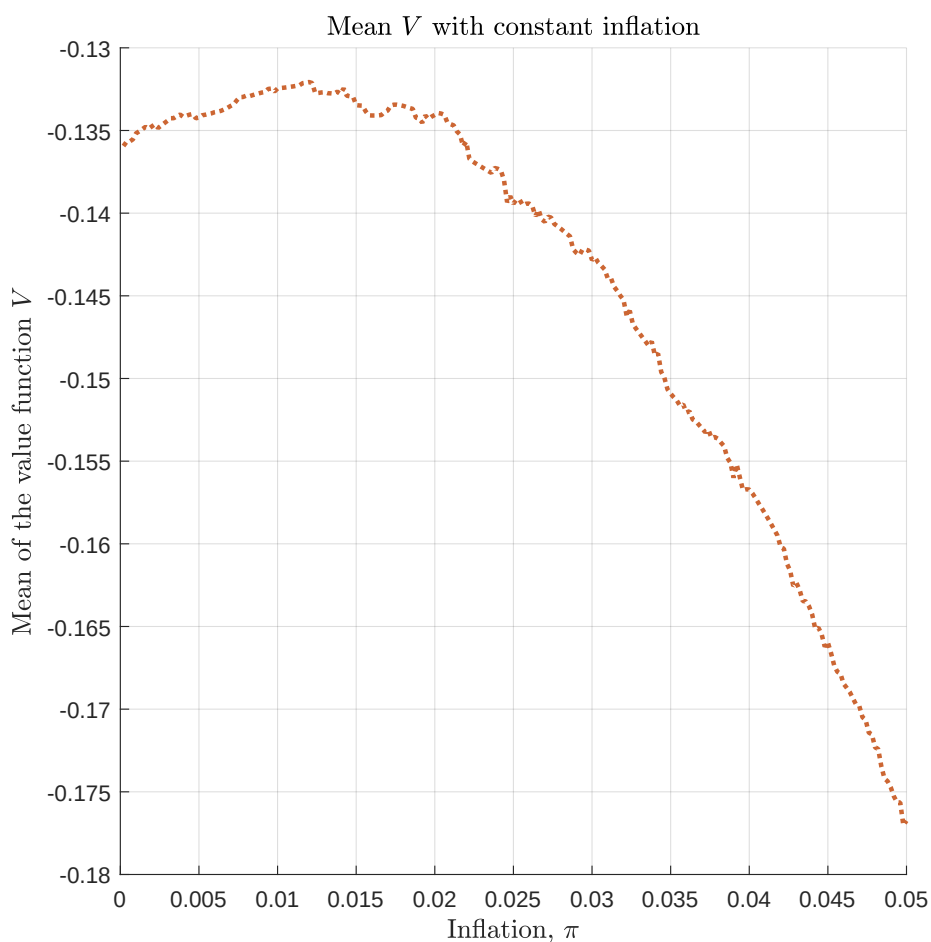


FIGURE D.3. Average welfare as a function of the value of inflation in a constant inflation rule. The dotted red line is the average level of inflation, computed using the ergodic distribution, as a function of the value of inflation in a constant inflation rule.

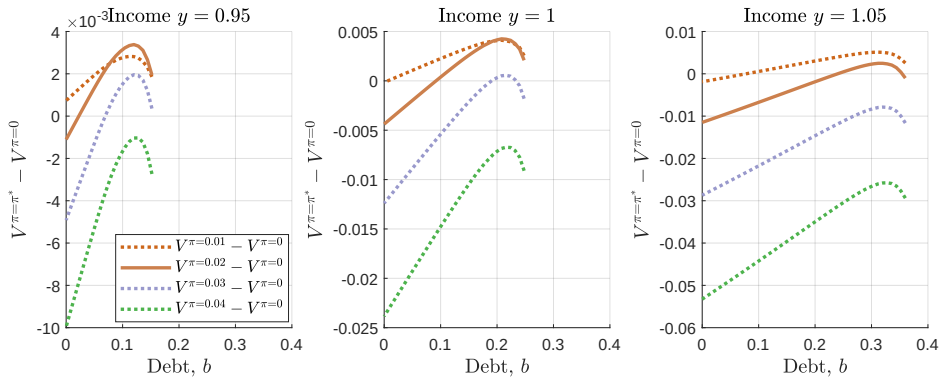


FIGURE D.4. Welfare as a function of the value of inflation in a constant inflation rule. The figure shows the difference between the value functions $V_{\pi}^{\pi=\pi^*} - V_{\pi}^{\pi=0}$ for different values of constant inflation π^* .